TANISHKA BAJPAI

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LinkedIn | GitHub | Portfolio

SUMMARY

Biomedical Engineering student specializing in Machine Intelligence, with experience in biomedical equipment maintenance, EEG signal processing, machine learning, research, and student-team leadership. Combines technical problem-solving with communication, event coordination, and cross-functional collaboration.

EDUCATION

Bachelor of Technology in Biomedical Engineering with Machine Intelligence **Expected May 2028**
SRM Institute of Science and Technology
- CGPA: 9.0

Class XII, PCB + IP **May 2023**
Vydehi School of Excellence
- Final Score: 86.8%
- Selected as Event Manager of the Student Council; organized interschool Model United Nations conferences.

Class X, Science **May 2021**
Vydehi School of Excellence
- Final Score: 93%

EXPERIENCE

Biomedical Engineering Intern **Jun 2026 – Jul 2026**
Sri Sathya Sai Institute of Higher Medical Sciences (SSSIHMS), Bengaluru
- Gained hands-on experience in calibration, troubleshooting, and preventive maintenance of biomedical equipment.
- Assisted with inspection and functional verification of devices across Radiology, Cardiology, Neurology, Laboratory Services, and Operation Theatres.
- Worked with CT, MRI, X-ray, ECG, EEG, ENMG, and automated laboratory analyzers.

R&D Associate Head **Oct 2025 – Present**
PLACFV's (SRM Student Placement Team), SRM Institute of Science and Technology
- Conducted industry and recruitment-trend research and compiled recruiter, HR outreach, and company insights.
- Led recruitment processes, coordinated volunteers across domains, and managed task delegation and cross-domain communication.

PROJECTS

EarSense: Development and Feasibility Evaluation of a Wearable Ear-Mounted EEG System for P300 Detection
Python, Biomedical Signal Processing, Machine Learning
- Investigating the feasibility of reduced-channel, ear-mounted EEG for wearable brain monitoring and P300 event-related potential detection.
- Analyzing EEG signals and evaluating signal reconstruction using classification performance and waveform correlation against conventional scalp EEG.
- Developing a machine learning pipeline to distinguish P300 and non-P300 responses, alongside hardware design for a compact, wearable EEG prototype.

Pan-Cancer Classification from RNA-Seq Gene Expression Using Self-Attention Neural Networks
Python, TensorFlow, SHAP, Flask
- Developed a self-attention neural network for multiclass cancer classification from RNA-Seq gene-expression data.
- Implemented SHAP-based explainability for biomarker discovery and model interpretation; built an interactive Flask application for prediction and visualization.

SKILLS

Biomedical & Research: Biomedical Signal Processing • Physiological Signal Analysis • Biomedical Instrumentation • Bioinformatics • Scientific Literature Review
AI, Machine Learning & Data Analysis: Machine Learning • Deep Learning / Neural Networks • Model Evaluation & Classification • Explainable AI (SHAP) • Statistical Analysis & Data Visualization
Programming & Analysis: Python • MATLAB • NumPy • Pandas • SciPy • Scikit-learn • TensorFlow • Flask • Excel
Professional: Team Leadership • Technical Writing • Research Presentation • Public Speaking • Event Coordination

ACCOMPLISHMENTS

- Second Runner-Up, PCB Design & Fabrication Workshop for Biomedical Applications (TekMedica, SRMIST).
- Presented a research poster at the International Conference on Biotechnology (ICB 2026).
- Published in an anthology in July 2021.

LANGUAGES

Hindi (First Language) • English (C2, Proficient) • French (A2, Elementary)